

YUNCONG YANG

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EDUCATION

University of Massachusetts, Amherst

Ph.D. student in Computer Science

Advisor: Chuang Gan

Research Interests: World Models, Embodied AI, Robotics

GPA: 3.95/4.00

Amherst, MA

Sep 2024 - Present

Columbia University

M.S. in Computer Science

Advisor: Shih-Fu Chang

Research Interests: Computer Vision, Multi-modal Foundation Models

New York, NY

Graduated Dec 2022

Columbia University

B.S. in Computer Science: Intelligent Systems Track

GPA: 4.05/4.33 (Top 5%) | Summa Cum Laude

New York, NY

Graduated May 2021

PUBLICATIONS

SyncWorld: Visual Calibration Enables World Models as Zero-Shot Simulators

NeurIPS 2026

[Oral Presentation, Top 0.3%]

Yuncong Yang*, Zhengtao Han*, Furkan Özyurt*, Zeyuan Yang, Han Yang, Junyi Cao, Haoyu Zhen, Yilun Du, Chuang Gan

Action Images: End-to-End Policy Learning via Multiview Video Generation

NeurIPS 2026

Haoyu Zhen, Zixian Gao, Qiao Sun, Yilin Zhao, **Yuncong Yang**, Yilun Du, Pengsheng Guo, Tsun-Hsuan Wang, Yi-Ling Qiao, Chuang Gan

Fast Spatial Memory with Elastic Test-Time Training

ECCV 2026

Ziqiao Ma, Xueyang Yu, Haoyu Zhen, **Yuncong Yang**, Joyce Chai, Chuang Gan

MindJourney: Test-Time Scaling with World Models for Spatial Reasoning

NeurIPS 2025

Yuncong Yang*, Jiageng Liu*, Zheyuan Zhang, Siyuan Zhou, Reuben Tan, Jianwei Yang, Yilun Du, Chuang Gan

Learning 3D Persistent Embodied World Models

NeurIPS 2025

Siyuan Zhou, Yilun Du, **Yuncong Yang**, Lei Han, Peihao Chen, Dit-Yan Yeung, Chuang Gan

3D-Mem: 3D Scene Memory for Embodied Exploration and Reasoning

CVPR 2025

Yuncong Yang*, Han Yang*, Jiachen Zhou, Peihao Chen, Hongxin Zhang, Yilun Du, Chuang Gan

TempCLR: Temporal Alignment Representation with Contrastive Learning

ICLR 2023

Yuncong Yang*, Jiawei Ma*, Shiyuan Huang, Long Chen, Xudong Lin, Guangxing Han, Shih-Fu Chang

MineDojo: Building Open-Ended Embodied Agents with Internet-Scale Knowledge

NeurIPS 2022

[Outstanding Paper Award, Featured Paper Presentation]

Linxi “Jim” Fan, Guanzhi Wang, Yunfan Jiang, Ajay Mandlekar, **Yuncong Yang**, Haoyi Zhu, Andrew Tang, De-An Huang, Yuke Zhu[†], Anima Anandkumar[†]

Few-Shot End-to-End Object Detection via Constantly Concentrated Encoding Across Heads

ECCV 2022

Jiawei Ma, Guangxing Han, Shiyuan Huang, **Yuncong Yang**, Shih-Fu Chang

RESEARCH EXPERIENCE

Meta

Research Intern | Advisor: Dr. Xitong Yang

May 2026 - Present

🌐 Embodied AI, World Models, and Robotics

- Developing in-context world models that predict the visual consequences of robot actions for tabletop manipulation.

University of Massachusetts, Amherst

Doctoral Research Assistant | Advisor: Prof. Chuang Gan

Sep 2024 - Present

🏠 World Models, Embodied AI, and 3D Scene Understanding

- Developed SyncWorld, a visually calibrated world model for zero-shot simulation across unseen cameras, environments, and robot embodiments.
- Developed 3D-Mem, a persistent 3D scene memory for lifelong exploration and spatial reasoning by embodied agents.

Microsoft

Research Intern | Advisors: Dr. Reuben Tan, Dr. Zhengyuan Yang

Jun 2025 - Dec 2025

🌐 Embodied AI and World Models

- Developed MindJourney, enabling VLMs to control a 3D-aware world model at test time for stronger spatial reasoning.
- Investigated active perception in 3D imagination for spatial reasoning and tabletop manipulation.

MIT-IBM Watson AI Lab

Research Collaborator | Advisor: Prof. Chuang Gan

Apr 2023 - Aug 2024

🏠 Foundation Models for Robotics

- Integrated Mixture-of-Experts into Decision Transformers to improve learning across heterogeneous robot datasets and observation spaces.

NVIDIA

Research Collaborator | Advisor: Dr. Jim Fan

Feb 2022 - Feb 2023

🏠 Embodied AI and LLM Agents

- Led MineDojo's task-suite design, contributed to the MineCLIP reward model, and built APIs and data pipelines used by Voyager.

Columbia University

Research Assistant | Advisor: Prof. Shih-Fu Chang

Dec 2021 - Dec 2022

🎬 Video Understanding

- Developed TempCLR, a sequence-level contrastive framework for video-text retrieval, action localization, and few-shot action recognition.

AWARDS & HONORS

- SyncWorld, *NeurIPS 2026 Oral (Top 0.3%)*
- Outstanding Paper Award, *NeurIPS 2022 Datasets and Benchmarks Track*
- Summa Cum Laude, *Columbia University*
- Student Member, *Phi Beta Kappa*
- Dean's List (8 semesters)

TEACHING EXPERIENCES

Foundations of Programming, University of Massachusetts, Amherst

Fall 2025

Introduction to Computation, University of Massachusetts, Amherst

Fall 2024

Computer Vision II, Columbia University

Spring 2022